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Software Project Management (BCO012A)

By:-

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Software Quality Management



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- Concerned with ensuring the required level of quality is achieved in a software product
- Involves the definition of appropriate quality standards and the definition of procedures to ensure that these standards are followed
- Works best when a 'quality culture' is created where quality is seen as everyone's responsibility



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Quality Definition

- Quality means that a product satisfies the demands of its specifications
- It also means achieving a high level of customer satisfaction with the product
- In software systems this is difficult
 - customer quality requirements (e.g. efficiency or reliability) often conflict with developer quality requirements (e.g. maintainability or reusability)
 - software specifications are often incomplete, inconsistent, or ambiguous



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Quality Management Activities

- Quality assurance
 - establishing organizational quality standards and procedures
- Quality planning
 - selecting and modifying applicable quality standards and procedures for a particular project
- Quality control
 - ensuring quality standards and procedures are followed by development team

Note: Quality management should be separated from project management to ensure independence.

Managing People In Software Environments

Main Concerns

4 main concerns:

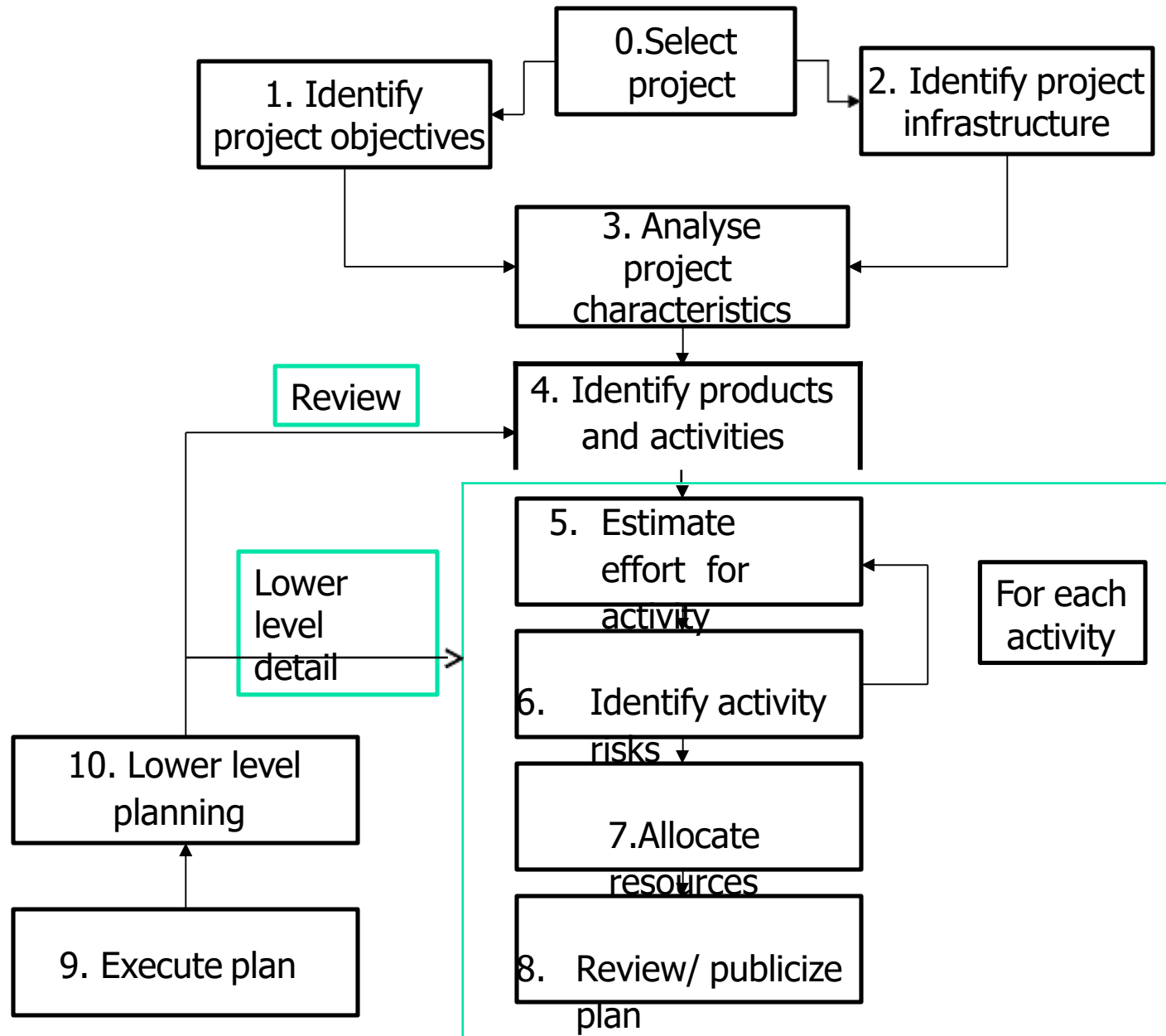
- Staff Selection
- Staff Development
- Staff Motivation
- Well-being Staff during course of project

‘Step Wise’ - an overview



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Understanding Behaviour

- Identify the handling of people
- 2 approach:
- Positivist approach
 - Based on development of system
 - Discipline of Organizational behaviour theories
 - “If A is the situation then B is likely to result”*



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- Interpretivist approach
 - How software Engineer defined?
 - Who customize and install package software
 - Cover the role of ICT business analyst
 - How successful defined?
- The 2 viewpoints(positivist and interpretivist)

are both valid and useful in managing the people in software field.



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- POSITIVIST APPROACH:
 - Tends to be objective and empirical
 - Seeks causes for behaviour
 - Conduct studies that can be generalised to larger populations
- INTERPRETIVIST METHODOLOGY
 - More qualitative
 - Based on smaller samples
 - View each consumption situation as unique and non-predictive
 - Look for common patterns across consumption situations



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Organizational Behaviour

- Fredrick Taylor attempted to analyse the most productive way of doing manual tasks.
- Taylor's three basic objectives
 - 1.To select the best man for the job.
 - 2.To instruct them in the best methods.
 - 3.To give incentives in the form of higher wages to the best workers.



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- The conditions under which the staff worked also affects productivity.
- OB researchers discovered that the state of the minds of the people influenced productivity.



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Theory X

- The average human has an innate dislike of work
- There is a need therefore for coercion, direction and control
- People tend to avoid responsibility



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Theory Y

- Work is as natural as rest or play
- External control and coercion are not the only ways of bringing about effort directed towards an organization's end
- Commitment to objectives is a function of the rewards associated with their achievement
- The average human can learn to accept and further seek responsibility
- The capacity to exercise imagination and other creative qualities is widely distributed.

Selecting the right person for the job

- Besides the s/w tools the individuals selected for a job affect the programming productivity.
- Experience
- Person who can communicate well.



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Recruitment Process

- Recruitment is often an organizational responsibility.
- Eligible candidates- have a curriculum vitae which shows the right and required details
- Suitable candidates- who can actually do the job well.
- Assesses actual skills rather than experience.



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- 1.Create a job specification.
- 2.Create a job holder profile.
- 3.Obtain applicants.
- 4.Examine CV's.
- 5.Interviews.
- 6.Other procedures.

Motivation

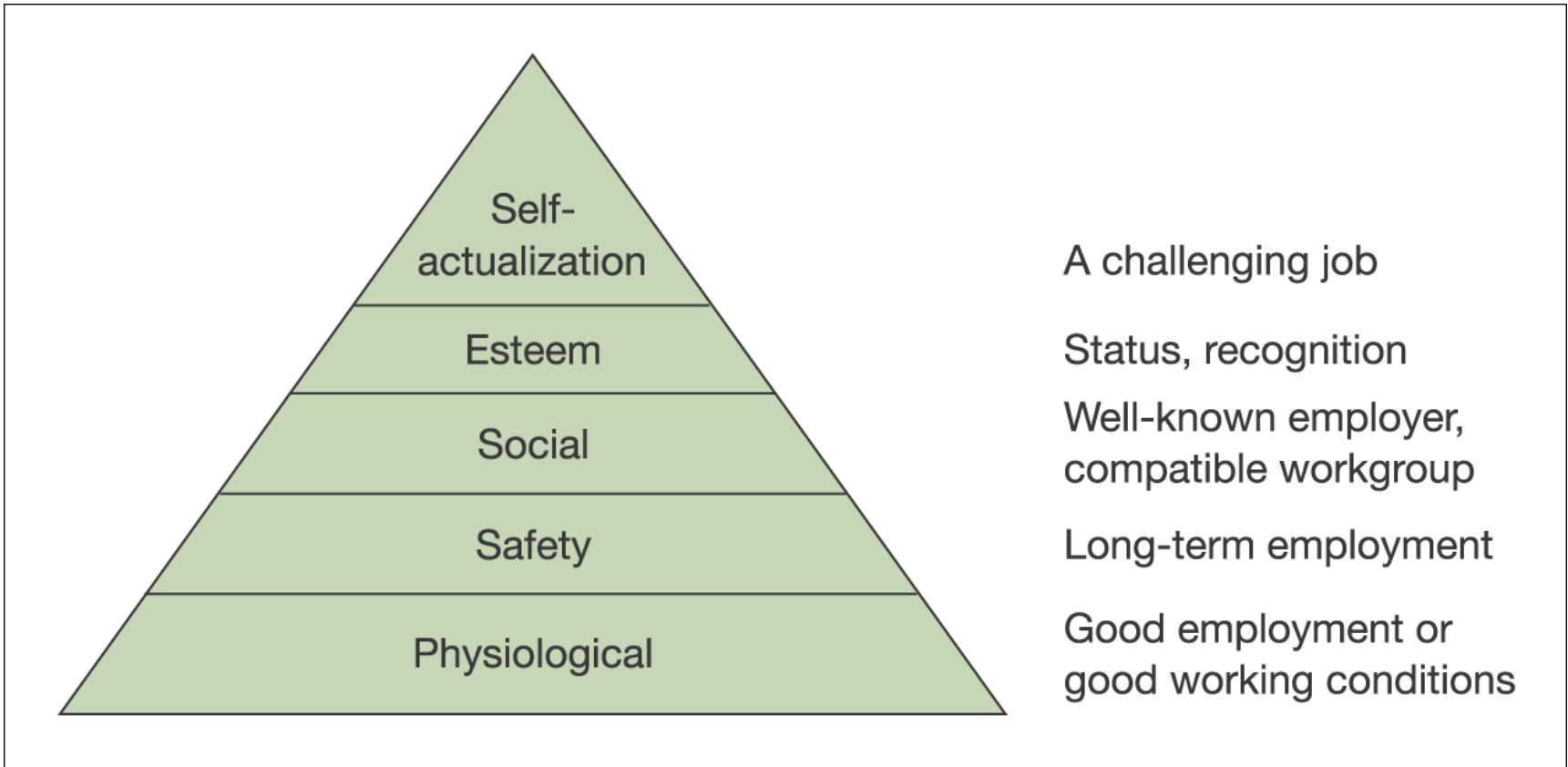
- Motivation and application can often make for shortfalls in innate skills
- Taylor's approach - financial incentives
- Abraham Maslow (1908-1970)
 - motivations vary from individual to individual. People will be motivated by different things at different times. People will always feel dissatisfied, but the focus of the dissatisfaction changes over time.
 - hierarchy of needs – as lower ones fulfilled, higher ones emerge

Maslow's hierarchy of needs



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Maslow's hierarchy of needs



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Herzberg

Herzberg suggested two sets of factors affected job satisfaction

Hygiene or maintenance factors – make you dissatisfied if they are not right e.g. pay, working conditions

Motivators – make you feel the job is worthwhile e.g. a sense of achievement or challenge



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Vroom

Vroom and colleagues identified three influences on motivation

Expectancy – the belief that working harder leads to better performance

Instrumentality – the belief that better performance will be rewarded

Perceived value of the reward

Note: If any of the above factors has a zero value, then motivation will also be zero

- Expectancy: - compilation error no matter how hard you tried
- Instrumentality – removing fault on the software that is no longer used
- Perceived value – 90% get 1st place.

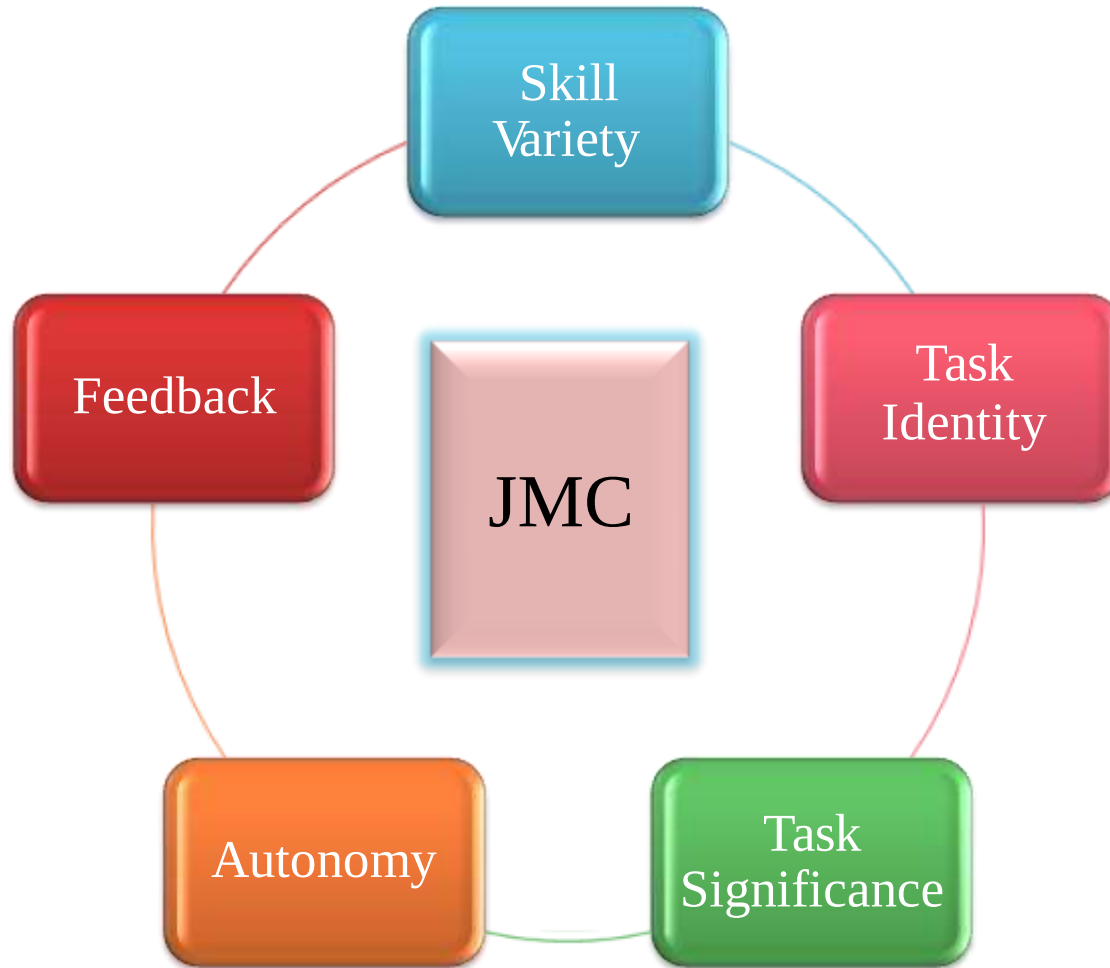
Oldham-Hackman Job Characteristics Model

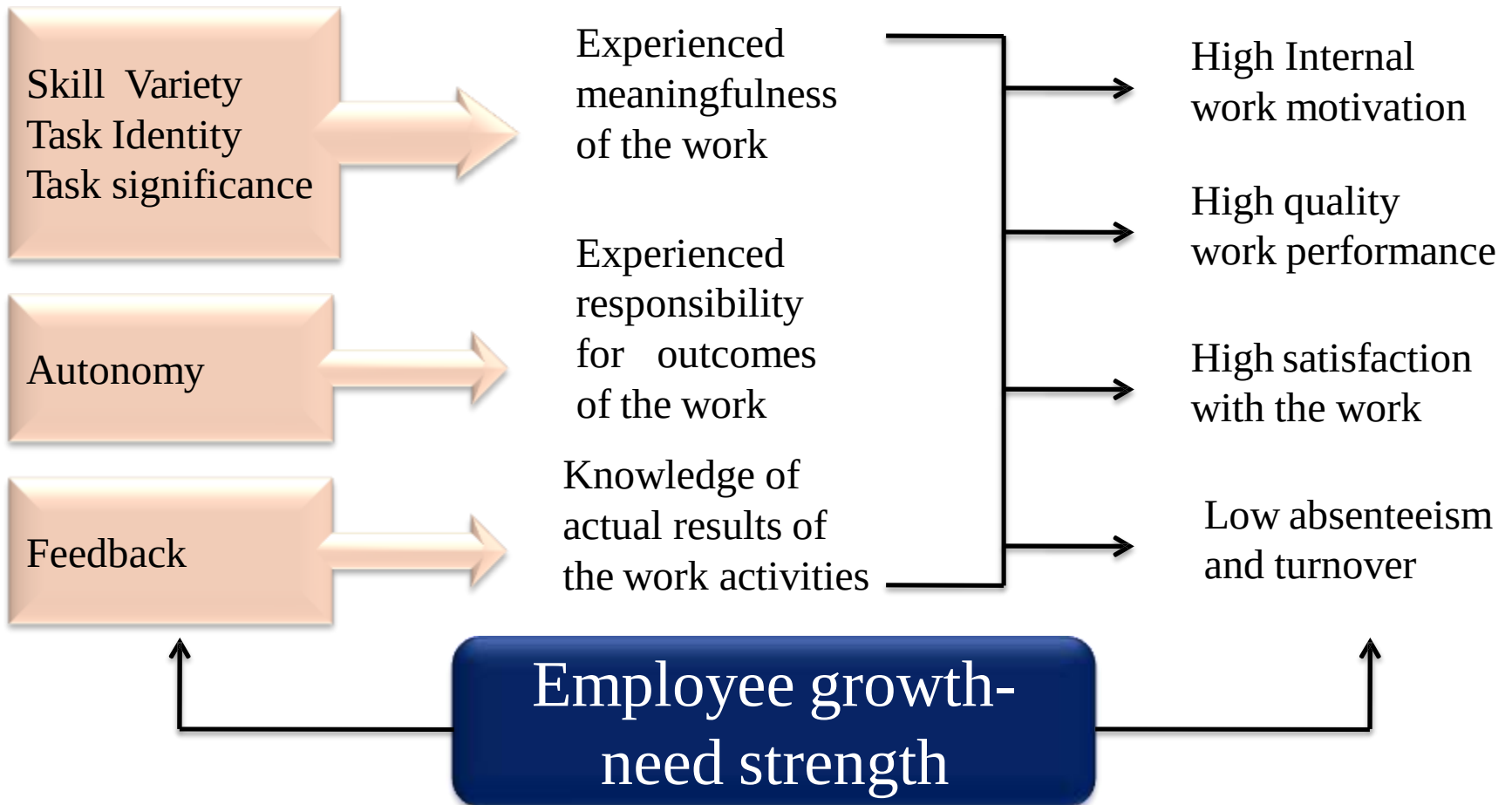
By J. Richard Hackman & Greg Oldham



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Methods to improve job satisfaction

- Set specific goals
- Provide feedback on the progress towards meeting those goals
- Consider job redesign
 - Job rotation
 - Job enlargement
 - Job enrichment



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Working in Teams

- Usually work/task will be divided among the groups.
- Formal Groups will be divided into subgroups
- There are two types
 - Command Group: Formal Mgmt Struct
 - Task Group

Step wise frame work



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- Identify project scope and objectives
- Identify project infrastructure
- Analyze project characteristics
- Estimate effort for each activity
- Identify activity risks
- Allocate resources
- Review/Publicize Plan



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Becoming a Team

- There are 5 basic stages of development
 - ▣ Forming:- Set up some **ground rules about behavior**
 - ▣ Storming:- **Conflicts arises**, everyone will try to dominate(Leadership).Group's methods of operation are established.
 - ▣ Norming:- **Conflicts are settled**. A feel of group identity emerges.
 - ▣ Performing:-The emphasis will be on the **task at hand**.
 - ▣ Adjourning:- The **group disbands**



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Management team roles

- The **co-ordinator** – good at chairing meetings
- The '**plant**' – an idea generator
- The **monitor-evaluator** – good at evaluating ideas
- The **shaper** – helps direct team's efforts
- The **team worker** – skilled at creating a good working environment

Belbin management roles - **continued**



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- The **resource investigator** – adept at finding resources, including information
- The **completer-finisher** – concerned with getting tasks completed
- The **implementer** – a good team player who is willing to undertake less attractive tasks if they are needed for team success
- The **specialist** – the 'techie' who likes to acquire knowledge for its own sake



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Group Performance

- One way of categorizing group tasks into:
 - Additive Tasks
 - Effort to **get final result**
 - Compensatory Tasks
 - **Judgments** of individual group members are pooled
 - Disjunctive Tasks
 - Only **one correct answer**
 - Conjunctive Tasks
 - Progress is governed by the **rate of slowest performer**



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Decision Making

- Decisions are categorized as
 - **Structured:** Simple, Routine Decisions (Rules can be applied in straight forward way)
 - **Unstructured:** More Complex and often requiring a degree of creativity

Decision Making Process



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Obstacles to make good decision

- Faulty Heuristics
 - “Rule of Thumb”
- Escalation of Commitment
 - Difficult to alter
- Information Overload
 - Too much of information



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Obstacles to good **GROUP** decision

- Groups deal less effectively with
 - ▣ Poor Structures
 - ▣ Problems need Creative Solutions
- Risky Shifts – Making decision on their own

Organization & Team Structure

- There are two important issues that are critical to the effective functioning of every organization are
 - Department Structure
 - How is a department is organized into teams?
 - Team Structure
 - How are project teams structured?

Department Structure



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- There are three broad ways in which a software development department can be structured:
 - ▣ Functional Format
 - ▣ Project Format
 - ▣ Matrix Format

Functional Format VS Project Format

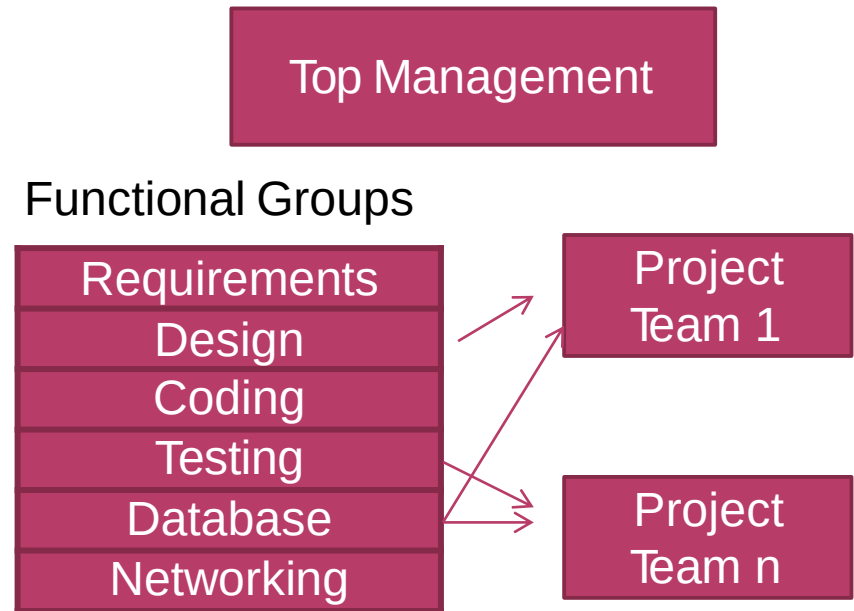


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a. Project Organization



b. Functional Organization

Matrix Format



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	Project			
Functional Group	#1	#2	#3	
#1	2	0	3	Functional Manager 1
#2	0	5	3	Functional Manager 2
#3	0	4	2	Functional Manager 3
#4	1	4	0	Functional Manager 4
#5	0	4	6	Functional Manager 5



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Team Structure

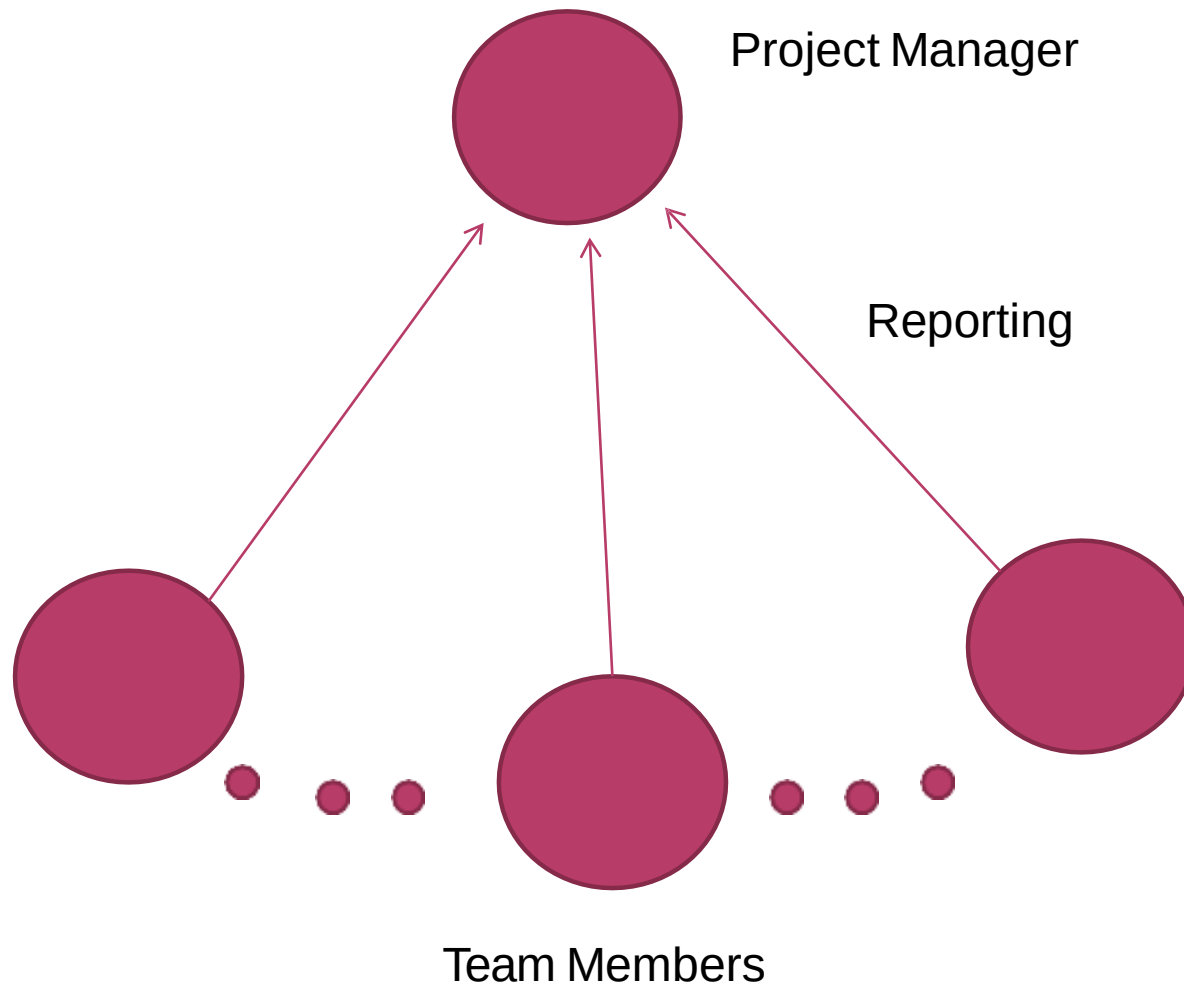
- Denotes the reporting, responsibility and communication structures in individual project teams
- There are only three team structures:
 - Democratic
 - Chief Programmer
 - Mixed team organization

Chief Programmer team



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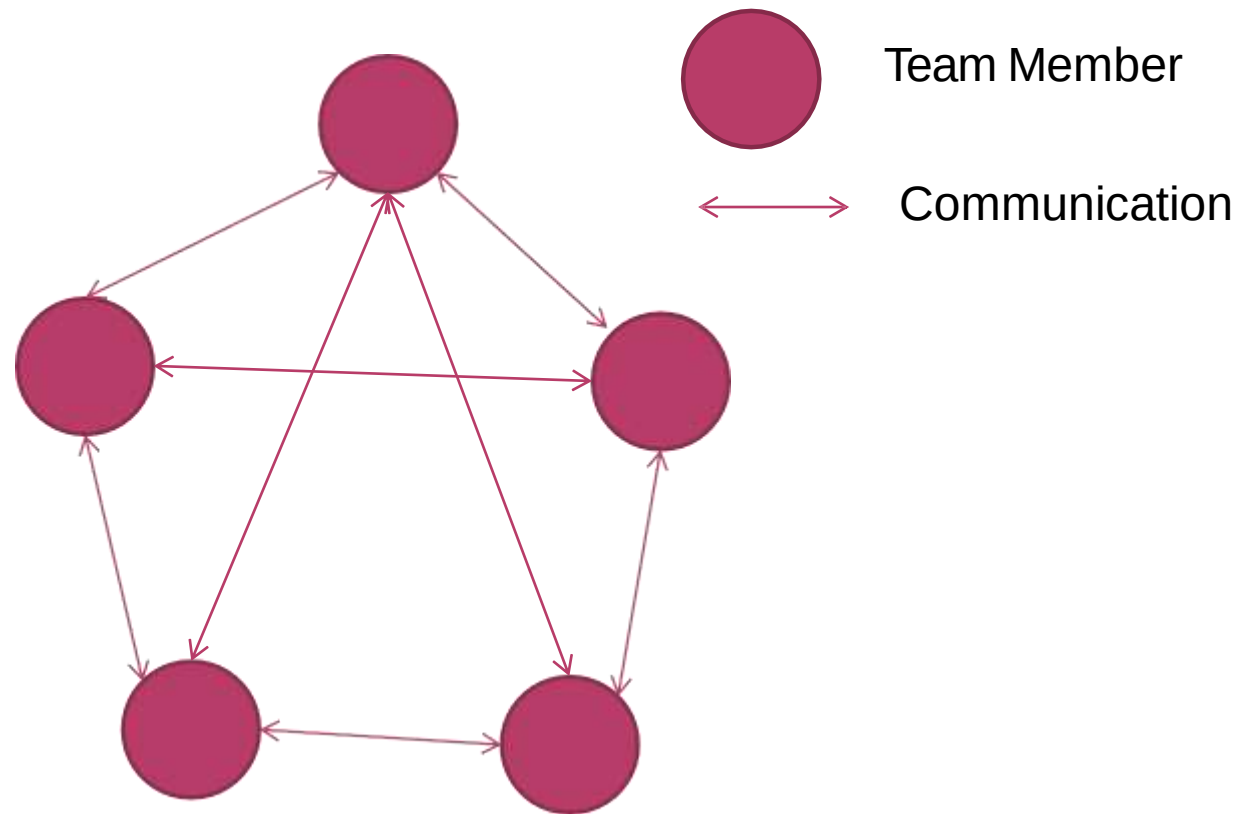


Democratic Team



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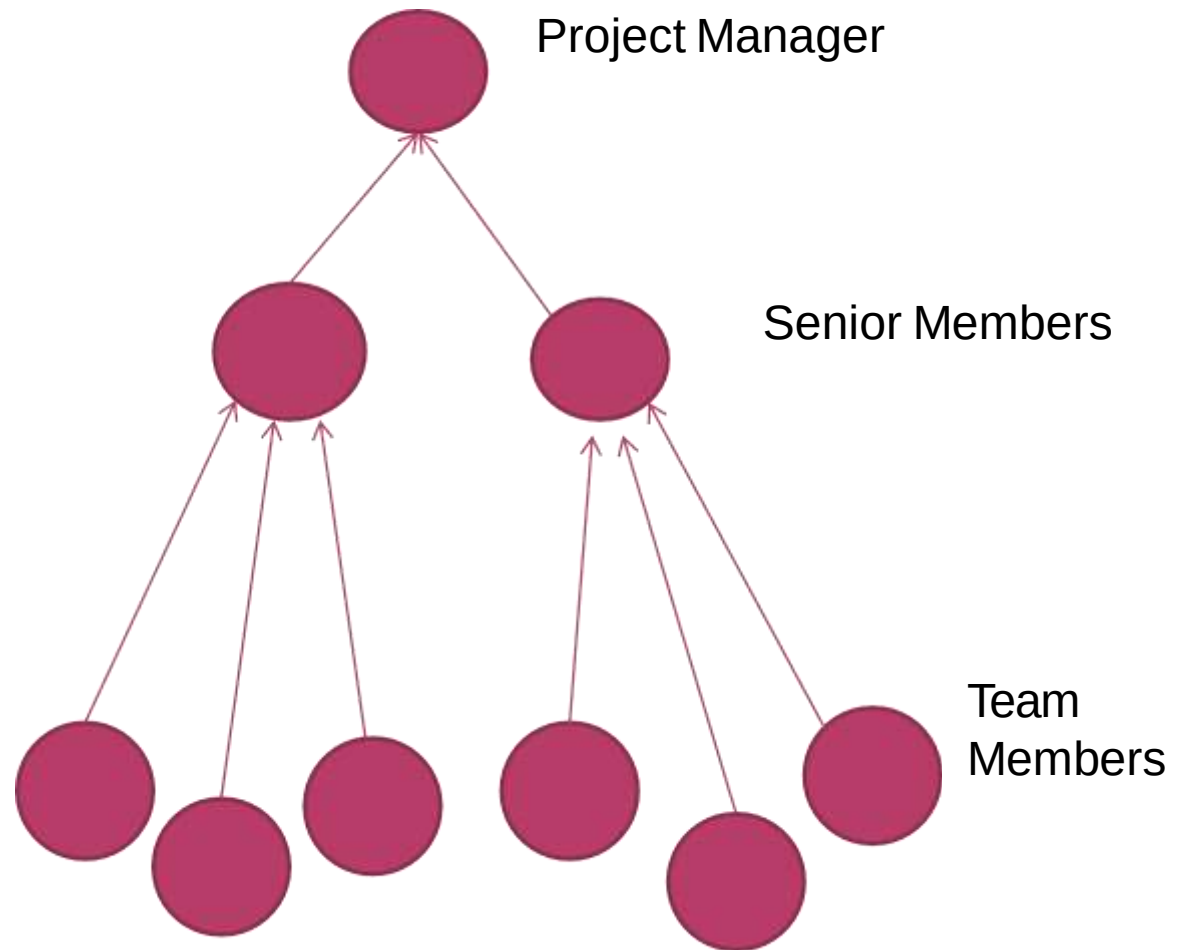


Mixed Control Team Structure



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Leadership

DEFINITION

- “ LEADERSHIP is the process of influencing the activities of an individual or a group in efforts towards goal achievement in a given situation”.
- It is the art of influencing and inspiring subordinates to perform their duties willingly, competently and enthusiastically for achievement of group objectives.



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QUALITIES OF LEADERSHIP

POSITIVE SELF IMAGE

CLEAR VISION

KEY LEADERSHIP
QUALITY

INNOVATIVE IDEAS TO
PROBLEM

QUICKLY SENSE
AND RESPOND TO
THE PROBLEM

IMPORTANCE OF LEADERSHIP



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- Securing Co-operation
- Use of power
- Co-ordination and command
- Maintaining Discipline
- Developing group morale



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Leadership: Types of authority

Position power

- Coercive power – able to threaten punishment
- Connection power – have access to those who do have power Legitimate
- power – based on a person's title conferring a special status
- Reward power – able to reward those who comply



Leadership: Types of power

Personal power

- *Expert power*: holder can carry out specialist tasks that are in demand
- *Information power*: holder has access to needed information
- *Referent power*: based on personal attractiveness or charisma

Leadership styles



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- Task orientation – focus on the work in hand
People orientation – focus on relationships
- Where there is uncertainty about the way job is to be done or staff are inexperienced they welcome task oriented supervision
- Uncertainty is reduced – people orientation more important

Risk that with reduction of uncertainty, managers have time on their hands and become more task oriented (interfering)

STYLES OF LEADERSHIP



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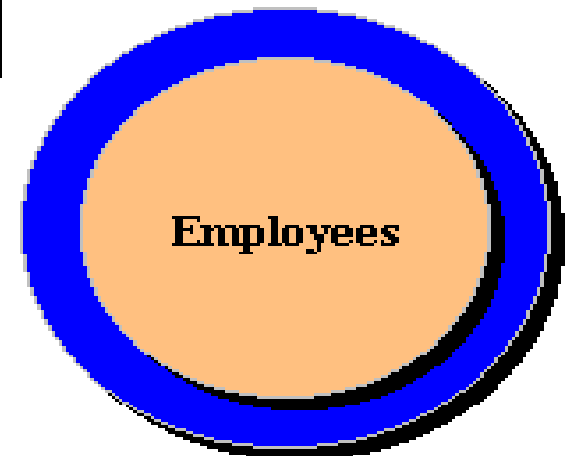
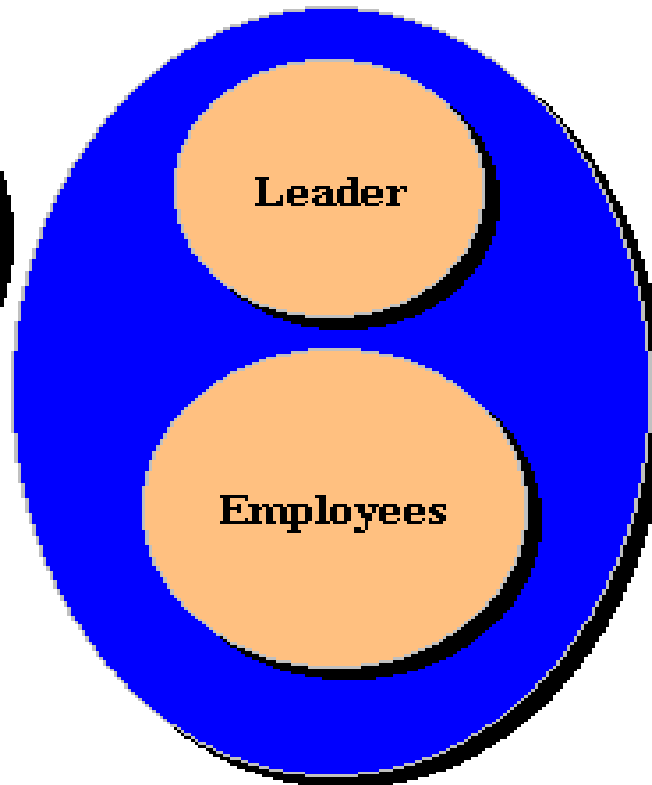
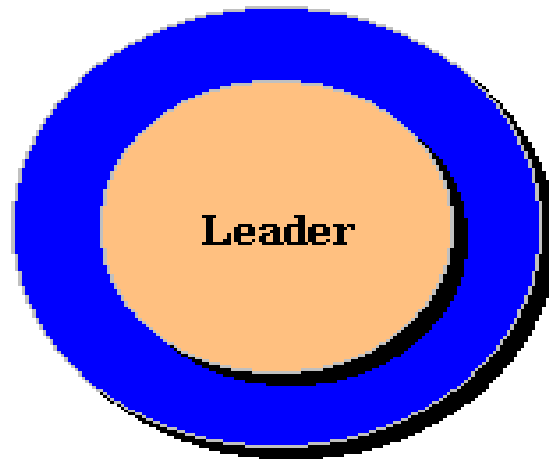
- AUTOCRATIC OR DICTATORIAL LEADERSHIP STYLE
- DEMOCRATIC OR PARTICIPATIVE LEADERSHIP STYLE
- LAISSEZ FAIRE OR FREE REIGN LEADERSHIP STYLE

POWER STYLE

Autocratic

Participative

Free-rein



Employees

Employees

Employees

Leader

Whole Group

Employees

EMPHASIS

AUTOCRATIC LEADERSHIP STYLE



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Autocratic

➤ **A leadership style where the leader makes all decisions independently or without consulting with subordinate**

Advantages:

- Provides strong motivation and reward for the leader
- Quick decision making takes place as single person decides for the whole group Subordinate are only to carry out the orders of the leader
- Decision making, planning or organizing need initiative
- Good in certain circumstances, such as urgent task or military action.

Disadvantages:

- poor decisions, poor level of employee motivation
- Leads to frustration, low morale and conflict among subordinate
- Subordinate tend to shirk responsibility and initiative.
- Full potential to subordinate and their creative ideas are not utilized
- Organizational continuity is threatened in the absence of the leader because subordinate get no opportunity for development

DEMOCRATIC LEADERSHIP STYLE



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-Democratic-

- **A leadership style where a leader encourages employee participation in decision-making**
- persuasive or consultative
- **Advantages:** better decisions, employee motivation
- It improves the job satisfaction and morale of subordinate
- The leader multiply his abilities through the contribution of his followers.
- It develops positive attitudes and reduces resistance to change
- **Disadvantages:** delayed decision, long consultation
- This style may not yield positive result when interaction of subordinate with leader is minimum.
- This may be used as passing the buck to others and of abdicating responsibility
- This style need tremendous communicating and persuasive skills on the part of leader.



LAISSEZ-FAIRE LEADERSHIP STYLE

Laissez-faire

- **A leadership style where employees are encouraged to make their own decisions within limits**
- **Advantages:** more freedom for employees
- Positive effect on job satisfaction and morale of subordinates
- Maximum possible scope for development of subordinates
- Full utilization of potential of subordinates
- **Disadvantages:** few guidelines, little incentive, poor motivation, maybe a mess
- Subordinate do not get the guidance and support of the leader
- It ignores the leader contribution just as Autocratic style ignores the contribution of the subordinates
- Subordinates may move in different directions and may work at cross purpose which may degenerate into chaos



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COMPARISON OF LEADERSHIP STYLE

	Autocratic Leadership	Democratic Leadership	Free-Rein Leadership
Concentration of Power	With Manager	Shared between Manager and Employees	With Employees — Minimal Supervision
Decision Making	By Manager	By Manager and Employees	By Employees
Communication Flow	From Top Down	Two Way	Widespread and as Needed

FACTORS AFFECTING LEADERSHIP STYLES



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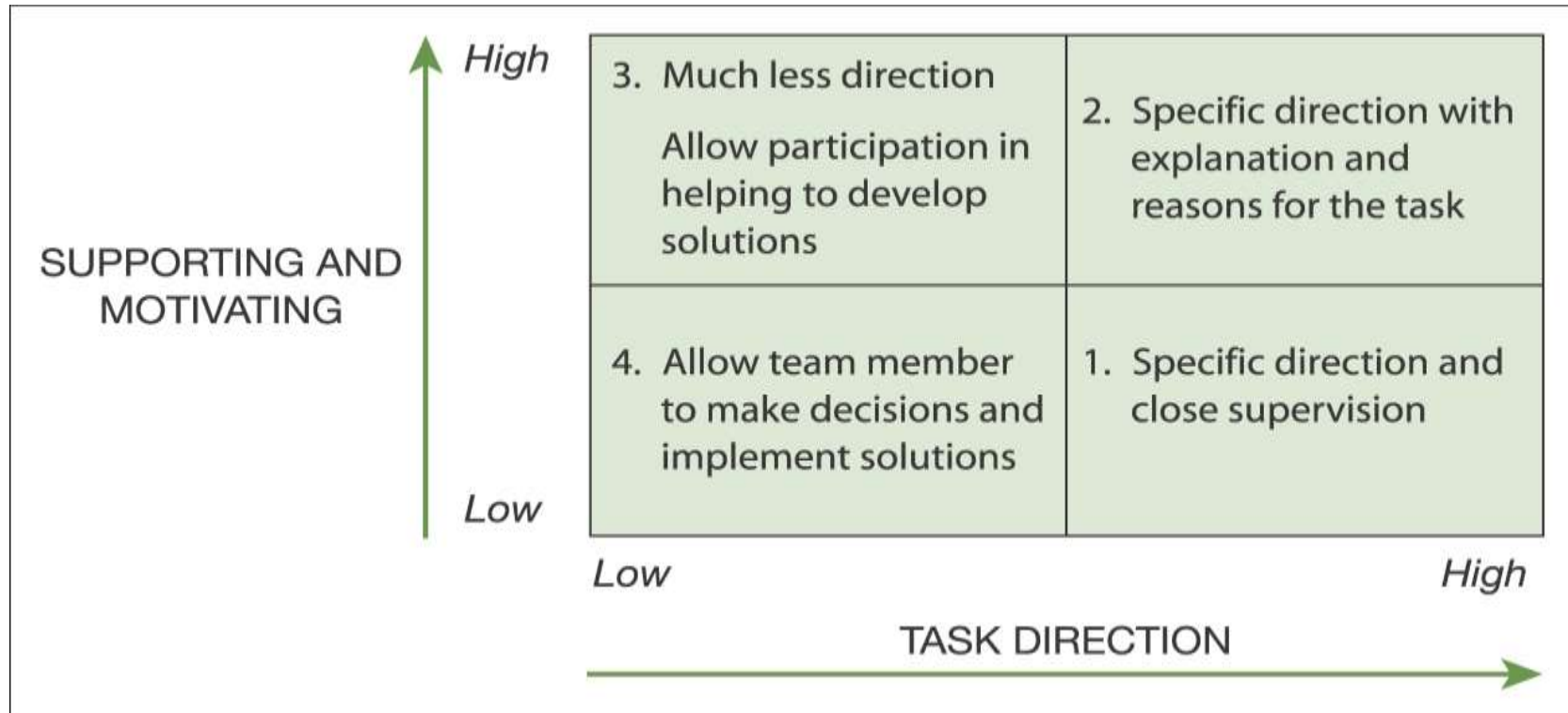
- The task
- The tradition of an organization
- The type of labor force
- The leader's personality
- The time
- Gender?

Coaching style matrix



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Developing team members by matching your supervision to their needs

Stress, Health & Safety Management



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- Design of work tasks
- Management styles
- Interpersonal relationships
- Workloads
- Career concerns
- Environmental conditions



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- Stress Management Programs
- Organizational change
- ✓ Workload in line with workers' capabilities and resources.
- ✓ Design jobs to provide meaning, stimulation, and opportunities
- Organizational change
- ✓ Clearly define roles and responsibilities
- ✓ Workers participate in decisions affecting their jobs
- ✓ Improved communications to reduce job stress
- ✓ Social interaction among workers
- ✓ Compatible work schedules

Others Techniques

Start a healthful life style

- Compartmentalize work/home life
- Get physical exercise
- Discuss problems with spouse or friends or peers at work
- Change to non-work activity
- Change strategy to work tasks
- Develop Positive Coping Skills



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Thank You